VISION TRANSFORMER

Image Classification Inference Pipeline

Lenovo Validated Design | Architecture draft

01 REQUEST & INPUT PREPARATION

1 Submit image
User sends JPEG / PNG
through the service API

2 Validate & decode
Check request and image*
Decode image to RGB

3 Preprocess image
Resize / crop / normalize
Create FP32 input tensor

4 Queue & batch
Collect prepared requests
Flush at size or time limit

02 ViT MODEL INFERENCE

PyTorch or OpenVINO | BF16 inference configuration

5 Patch embedding
Split image into patches
Project patches into vectors

6 Token sequence
Add a classification token
Add positional embeddings

7 Encoder layers × L
Multi-head self-attention
MLP, norms & residuals

8 Classification head
Read classification token
Produce class scores

03 POSTPROCESSING & RESPONSE

9 Select predictions
Benchmark: argmax → top-1 class
Service option: top-k / softmax

10 Build response
Map class IDs to label names
Serialize labels and scores*

11 Return to user
Send classification results
through the service API*

Scope: standard ViT classifier with a CLS token; ResNet and Swin use different model internals.

* API handling, request validation and response formatting are proposed service stages, not benchmark-validated features.

Precision note: FP32 preprocessing / OpenVINO input is compatible with BF16 inference configuration.